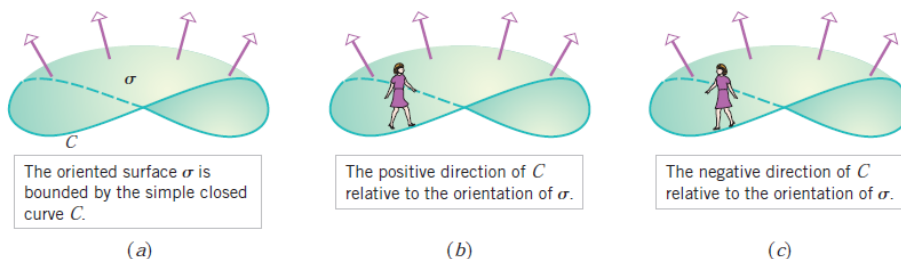


15.8 STOKES' THEOREM

In this section we will discuss a generalization of Green's Theorem to three dimensions that has important applications in the study of vector fields, particularly in the analysis of rotational motion of fluids. This theorem will also provide us with a physical interpretation of the curl of a vector field.



▲ Figure 15.8.1

Curl

In earlier section, we defined the ~~divergence~~ of a vector field

$$\mathbf{F}(x, y, z) = f(x, y, z)\mathbf{i} + g(x, y, z)\mathbf{j} + h(x, y, z)\mathbf{k}$$

as

$$\text{curl } \mathbf{F} = \left(\frac{\partial h}{\partial y} - \frac{\partial g}{\partial z} \right) \mathbf{i} + \left(\frac{\partial f}{\partial z} - \frac{\partial h}{\partial x} \right) \mathbf{j} + \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) \mathbf{k} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f & g & h \end{vmatrix} \quad (1)$$

but we did not attempt to give a physical explanation of its meaning at that time. The following result, known as **Stokes' Theorem**, will provide us with a physical interpretation of the curl in the context of fluid flow.

15.8.1 THEOREM (Stokes' Theorem) Let σ be a piecewise smooth oriented surface that is bounded by a simple, closed, piecewise smooth curve C with positive orientation. If the components of the vector field

$$\mathbf{F}(x, y, z) = f(x, y, z)\mathbf{i} + g(x, y, z)\mathbf{j} + h(x, y, z)\mathbf{k}$$

are continuous and have continuous first partial derivatives on some open set containing σ , and if $\mathbf{T}$ is the unit tangent vector to C , then

$$\oint_C \mathbf{F} \cdot \mathbf{T} \, ds = \iint_{\sigma} (\text{curl } \mathbf{F}) \cdot \mathbf{n} \, dS \quad (2)$$

If $\mathbf{F}$ is a force field, the integral on the left side of (2) represents the work performed by the force field on a particle that traverses the curve C . Thus, loosely phrased, Stokes' Theorem states:

The work performed by a force field on a particle that traverses a simple, closed, piecewise smooth curve C in the positive direction can be obtained by integrating the normal component of the curl over an oriented surface σ bounded by C .

■ USING STOKES' THEOREM TO CALCULATE WORK

For computational purposes it is usually preferable to use Formula (30) in Section 15.2 to rewrite the formula in Stokes' Theorem as

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_{\sigma} (\text{curl } \mathbf{F}) \cdot \mathbf{n} \, dS \quad (3)$$

Stokes' Theorem is usually the method of choice for calculating work around piecewise smooth curves with multiple sections, since it eliminates the need for a separate integral evaluation over each section. This is illustrated in the following example.

Note from Section 15.6:

15.6.3 THEOREM Let σ be a smooth surface of the form $z = g(x, y)$, $y = g(z, x)$, or $x = g(y, z)$, and suppose that the component functions of the vector field $\mathbf{F}$ are continuous on σ . Suppose also that the equation for σ is rewritten as $G(x, y, z) = 0$ by taking g to the left side of the equation, and let R be the projection of σ on the coordinate plane determined by the independent variables of g . If σ has positive orientation, then

$$\Phi = \iint_{\sigma} \mathbf{F} \cdot \mathbf{n} \, dS = \iint_R \mathbf{F} \cdot \nabla G \, dA \quad (11)$$

Formula (11) can either be used directly for computations or to derive some more specific formulas for each of the three surface types. For example, if $z = g(x, y)$, then we have $G(x, y, z) = z - g(x, y)$, so

$$\nabla G = -\frac{\partial g}{\partial x} \mathbf{i} - \frac{\partial g}{\partial y} \mathbf{j} + \mathbf{k} = -\frac{\partial z}{\partial x} \mathbf{i} - \frac{\partial z}{\partial y} \mathbf{j} + \mathbf{k}$$

Substituting this expression for ∇G in (11) and taking R to be the projection of the surface $z = g(x, y)$ on the xy -plane yields

$$\iint_{\sigma} \mathbf{F} \cdot \mathbf{n} \, dS = \iint_R \mathbf{F} \cdot \left(-\frac{\partial z}{\partial x} \mathbf{i} - \frac{\partial z}{\partial y} \mathbf{j} + \mathbf{k} \right) dA \quad \boxed{\sigma \text{ of the form } z = g(x, y) \text{ and oriented up}} \quad (12)$$

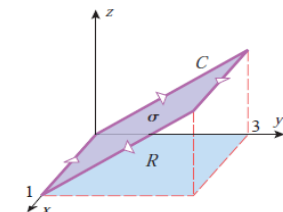
$$\iint_{\sigma} \mathbf{F} \cdot \mathbf{n} \, dS = \iint_R \mathbf{F} \cdot \left(\frac{\partial z}{\partial x} \mathbf{i} + \frac{\partial z}{\partial y} \mathbf{j} - \mathbf{k} \right) dA \quad \boxed{\sigma \text{ of the form } z = g(x, y) \text{ and oriented down}} \quad (13)$$

The derivations of the corresponding formulas when $y = g(z, x)$ and $x = g(y, z)$ are left as exercises.

► **Example 1** Find the work performed by the force field

$$\mathbf{F}(x, y, z) = x^2\mathbf{i} + 4xy^3\mathbf{j} + y^2x\mathbf{k}$$

on a particle that traverses the rectangle C in the plane $z = y$ shown in Figure 15.8.2.



▲ Figure 15.8.2

Solution. The work performed by the field is

$$W = \oint_C \mathbf{F} \cdot d\mathbf{r}$$

However, to evaluate this integral directly would require four separate integrations, one over each side of the rectangle. Instead, we will use Formula (3) to express the work as the surface integral

$$W = \iint_{\sigma} (\text{curl } \mathbf{F}) \cdot \mathbf{n} \, dS$$

in which the plane surface σ enclosed by C is assigned a *downward* orientation to make the orientation of C positive, as required by Stokes' Theorem.

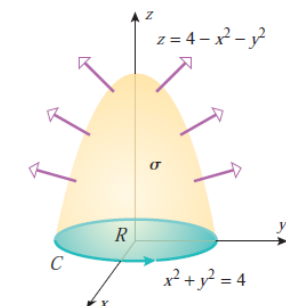
Since the surface σ has equation $z = y$ and

$$\text{curl } \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 & 4xy^3 & xy^2 \end{vmatrix} = 2xy\mathbf{i} - y^2\mathbf{j} + 4y^3\mathbf{k}$$

it follows from Formula (13) of Section 15.6 with $\text{curl } \mathbf{F}$ replacing $\mathbf{F}$ that

$$\begin{aligned} W &= \iint_{\sigma} (\text{curl } \mathbf{F}) \cdot \mathbf{n} \, dS = \iint_R (\text{curl } \mathbf{F}) \cdot \left(\frac{\partial z}{\partial x} \mathbf{i} + \frac{\partial z}{\partial y} \mathbf{j} - \mathbf{k} \right) dA \\ &= \iint_R (2xy\mathbf{i} - y^2\mathbf{j} + 4y^3\mathbf{k}) \cdot (0\mathbf{i} + \mathbf{j} - \mathbf{k}) \, dA \\ &= \int_0^1 \int_0^3 (-y^2 - 4y^3) \, dy \, dx \\ &= - \int_0^1 \left[\frac{y^3}{3} + y^4 \right]_{y=0}^3 \, dx \\ &= - \int_0^1 90 \, dx = -90 \quad \blacktriangleleft \end{aligned}$$

► **Example 2** Verify Stokes' Theorem for the vector field $\mathbf{F}(x, y, z) = 2z\mathbf{i} + 3x\mathbf{j} + 5y\mathbf{k}$, taking σ to be the portion of the paraboloid $z = 4 - x^2 - y^2$ for which $z \geq 0$ with upward orientation, and C to be the positively oriented circle $x^2 + y^2 = 4$ that forms the boundary of σ in the xy -plane (Figure 15.8.3).



▲ Figure 15.8.3

Solution. We will verify Formula (3). Since σ is oriented up, the positive orientation of C is counterclockwise looking down the positive z -axis. Thus, C can be represented parametrically (with positive orientation) by

$$x = 2 \cos t, \quad y = 2 \sin t, \quad z = 0 \quad (0 \leq t \leq 2\pi) \quad (4)$$

Therefore,

$$\mathbf{f} = 2z \mathbf{i} + 3x \mathbf{j} + 5y \mathbf{k}$$

$$d\mathbf{r} = dx \mathbf{i} + dy \mathbf{j} + dz \mathbf{k}$$

$$\begin{aligned} \oint_C \mathbf{F} \cdot d\mathbf{r} &= \oint_C 2z \, dx + 3x \, dy + 5y \, dz \\ &= \int_0^{2\pi} [0 + (6 \cos t)(2 \cos t) + 0] \, dt \\ &= \int_0^{2\pi} 12 \cos^2 t \, dt = 12 \left[\frac{1}{2}t + \frac{1}{4} \sin 2t \right]_0^{2\pi} = 12\pi \end{aligned}$$

To evaluate the right side of (3), we start by finding $\text{curl } \mathbf{F}$. We obtain

$$\text{curl } \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2z & 3x & 5y \end{vmatrix} = 5\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$$

Since σ is oriented up and is expressed in the form $z = g(x, y) = 4 - x^2 - y^2$, it follows

from Formula (12) of Section 15.6 with $\text{curl } \mathbf{F}$ replacing $\mathbf{F}$ that

$$\begin{aligned}
 \iint_{\sigma} (\text{curl } \mathbf{F}) \cdot \mathbf{n} \, dS &= \iint_R (\text{curl } \mathbf{F}) \cdot \left(-\frac{\partial z}{\partial x} \mathbf{i} - \frac{\partial z}{\partial y} \mathbf{j} + \mathbf{k} \right) dA \\
 &= \iint_R (5\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}) \cdot (2x\mathbf{i} + 2y\mathbf{j} + \mathbf{k}) dA \\
 &= \iint_R (10x + 4y + 3) dA \\
 &= \int_0^{2\pi} \int_0^2 (10r \cos \theta + 4r \sin \theta + 3)r \, dr \, d\theta \\
 &= \int_0^{2\pi} \left[\frac{10r^3}{3} \cos \theta + \frac{4r^3}{3} \sin \theta + \frac{3r^2}{2} \right]_{r=0}^2 d\theta \\
 &= \int_0^{2\pi} \left(\frac{80}{3} \cos \theta + \frac{32}{3} \sin \theta + 6 \right) d\theta \\
 &= \left[\frac{80}{3} \sin \theta - \frac{32}{3} \cos \theta + 6\theta \right]_0^{2\pi} = 12\pi
 \end{aligned}$$

Remark-1:

As guaranteed by Stokes' Theorem, the value of this surface integral is the same as the value obtained for the line integral. Note, however, that the line integral was simpler to evaluate and hence would be the method of choice in this case.

Remark-2:

Observe that in Formula (3) the only relationships required between σ and C are that C be the boundary of σ and that C be positively oriented relative to the orientation of σ . Thus, if σ_1 and σ_2 are *different* oriented surfaces but have the *same* positively oriented boundary curve C , then it follows from (3) that

$$\iint_{\sigma_1} \text{curl } \mathbf{F} \cdot \mathbf{n} \, dS = \iint_{\sigma_2} \text{curl } \mathbf{F} \cdot \mathbf{n} \, dS$$

For example, the parabolic surface in Example 2 has the same positively oriented boundary C as the disk R in Figure 15.8.3 with upper orientation. Thus, the value of the surface integral in that example would not change if σ is replaced by R (or by any other oriented surface that has the positively oriented circle C as its boundary). This can be useful in computations because it is sometimes possible to circumvent a difficult integration by changing the surface of integration.

■ RELATIONSHIP BETWEEN GREEN'S THEOREM AND STOKES' THEOREM

It is sometimes convenient to regard a vector field

$$\mathbf{F}(x, y) = f(x, y)\mathbf{i} + g(x, y)\mathbf{j}$$

in 2-space as a vector field in 3-space by expressing it as

$$\mathbf{F}(x, y) = f(x, y)\mathbf{i} + g(x, y)\mathbf{j} + 0\mathbf{k} \quad (5)$$

If R is a region in the xy -plane enclosed by a simple, closed, piecewise smooth curve C , then we can treat R as a *flat* surface, and we can treat a surface integral over R as an ordinary double integral over R . Thus, if we orient R up and C counterclockwise looking down the positive z -axis, then Formula (3) applied to (5) yields

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_R \text{curl } \mathbf{F} \cdot \mathbf{k} \, dA \quad (6)$$

But

$$\text{curl } \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f & g & 0 \end{vmatrix} = -\frac{\partial g}{\partial z}\mathbf{i} + \frac{\partial f}{\partial z}\mathbf{j} + \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y}\right)\mathbf{k} = \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y}\right)\mathbf{k}$$

since $\partial g/\partial z = \partial f/\partial z = 0$. Substituting this expression in (6) and expressing the integrals in terms of components yields

$$\oint_C f \, dx + g \, dy = \iint_R \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y}\right) \, dA$$

which is Green's Theorem [Formula (1) of Section 15.4]. Thus, we have shown that Green's Theorem can be viewed as a special case of Stokes' Theorem.

Practice Problem:

Chapter-15.8:

Problems: 5, 8--12.

Thank you for your attention!